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Enhancing creative learning in Thai classical dance education: An integration of digital preservation framework with ubiquitous learning environment☆

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ABSTRACT

The preservation of Thai classical dance as intangible cultural heritage faces critical challenges in documenting complex movements while maintaining cultural authenticity. Through a three-phase methodology comprising investigation, design, and evaluation, we developed a learning environment model with three key components: Creative Resource Hub for archival documentation, Interactive Studio with movement analysis capabilities, and Collaborative Workshop for preserving teaching methodologies. Testing with 78 secondary school students (39 experimental, 39 control) using cluster random sampling and validated Creative Thinking Assessment Rubric showed significant improvements in creative thinking skills in the experimental group compared to control (mean difference = 6.80, 95 % CI [5.10, 8.50], $p < 0.001$, Cohen's $d = 1.72$). The integration of creative development approaches with digital preservation technologies effectively captures both tangible movements and intangible cultural elements while enhancing accessibility through ubiquitous learning features, offering a sustainable model for comprehensive cultural heritage preservation.

1. Introduction

Over the past two decades, significant advances have been made in the field of dance digitization and documentation, yet comprehensive preservation solutions remain elusive (Stavrakis et al., 2012; Kitsikidis et al., 2015). While pioneering work has established fundamental methods for capturing traditional dance movements, current approaches often fail to address the full complexity of cultural transmission. Recent comprehensive reviews by Aristidou et al. (2019, 2022) and Chantas et al. (2018) have documented both the evolution of digital dance preservation methods and persistent challenges in maintaining cultural authenticity. Traditional dance embodies not only physical movements but also rich cultural knowledge, teaching practices, and artistic expressions that reflect the heritage of communities over generations. However, the growing loss of cultural knowledge due to globalization, urbanization, and generational shifts underscores the urgency of preserving these intangible elements. Without innovative technological interventions, there is a risk that critical aspects of traditional dance could be lost forever, depriving future generations of their cultural heritage.

This urgency has prompted researchers and practitioners to explore new approaches that integrate technology with traditional cultural practices. The preservation and development of traditional dance forms in the digital age presents a unique challenge in safeguarding intangible cultural heritage, particularly traditional dance forms, which faces unprecedented difficulties (Ami-Williams et al., 2024; Buragohain et al.,

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2024; Lyu, 2024). Traditional dance represents a complex form of cultural heritage that encompasses not only physical movements but also embedded cultural knowledge, teaching methodologies, and artistic expressions that have been transmitted through generations. For instance, Ami-Williams et al. (2024) explored the complexities of digitizing the Eyo masquerade dance, a Nigerian cultural performance, where the attire is intricately layered, affecting motion capture and requiring sophisticated simulation techniques to preserve its authenticity effectively. These findings highlight the necessity of balancing technological innovations with cultural sensitivity to ensure that digital preservation efforts align with the principles of authenticity and heritage conservation (Yatsunami, 2024; Prasad et al., 2024). As digital technologies reshape cultural preservation paradigms, there is an urgent need to develop innovative methods for documenting and transmitting these cultural practices while maintaining their authenticity and integrity (Yatsunami, 2024; Prasad et al., 2024). In the Eyo case study, Ami-Williams et al. (2024) illustrate the importance of using both modeling and physics-based simulation for costumes, demonstrating the necessity of balancing traditional cultural elements with technological innovation. This example highlights the intricate balance between technological innovation and cultural sensitivity, a balance that is crucial for successful digital preservation efforts.

Recent research in cultural heritage preservation has identified significant challenges in implementing digital documentation technologies for traditional dance forms. Studies by Dania et al. (2011) and Parrish (2007) highlight critical barriers in digital preservation efforts, including technological limitations in capturing complex movement patterns and challenges in preserving the nuanced aspects of traditional teaching methodologies. Ami-Williams et al. (2024) also discuss how motion capture systems may fail to capture precise movements due to the constraints posed by heavy traditional costumes, necessitating advanced methodologies for both movement and attire preservation. These challenges are particularly significant in classical dance preservation, where the integration of modern documentation methods must carefully balance innovation with cultural authenticity. These challenges underline the need for innovative solutions that address both the technical and cultural dimensions of dance preservation. The integration of these dimensions forms the foundation for comprehensive and culturally sensitive preservation frameworks.

The digital preservation of traditional dance faces additional complexities in capturing and transmitting creative processes. While advanced technologies such as virtual reality and motion capture systems offer new possibilities for cultural heritage documentation (Zhao, 2022; Li et al., 2024), many cultural institutions struggle to effectively implement these tools for comprehensive preservation purposes. Ami--Williams et al. (2024) emphasize that traditional dance preservation must consider multiple dimensions, including movement patterns, teaching methodologies, and cultural contexts, highlighting the integration of metadata to document the dance's sociocultural aspects and ensuring that costume and ritual details are preserved. To address these challenges, it is essential to explore innovative frameworks that incorporate both technical precision and cultural adaptability.

The integration of creativity in classical dance preservation serves several essential functions. First, creative thinking enables students to deeply understand and internalize traditional forms while developing their artistic capabilities. Second, creativity in teaching methodologies helps preserve the dynamic nature of dance traditions, ensuring they remain relevant while maintaining authenticity. Third, creative approaches to documentation allow for more comprehensive preservation of both physical movements and cultural context. In this way, creativity bridges traditional practices with contemporary pedagogical strategies, fostering both innovation and continuity in cultural transmission.

The ubiquitous learning environment in our framework serves multiple stakeholders in dance preservation. Dance students benefit from continuous access to learning resources and tools that support both traditional practice and creative development. Teachers gain new methods for transmitting traditional knowledge while adapting to modern educational needs. Cultural institutions benefit from improved documentation and preservation capabilities. This approach particularly benefits remote communities and international students seeking to learn traditional dance forms, addressing accessibility challenges in cultural heritage education. The inclusion of ubiquitous learning ensures that the preservation of traditional dances extends beyond documentation, enabling dynamic learning and interaction for diverse audiences.

Current approaches to digital cultural heritage preservation have primarily focused on basic documentation rather than comprehensive preservation solutions. Although studies by Karkou et al. (2008) and Mabingo (2015) demonstrate the potential of digital platforms in cultural heritage preservation, there remains a critical need for integrated preservation systems that can capture both tangible movements and intangible cultural elements. The 4 E theoretical framework (Embodied, Embedded, Enactive, and Extended Cognition) introduced by Griffith (2021) suggests promising directions for comprehensive digital preservation, yet practical implementations remain limited. Recent work by Weber and Reed (2020) validates the importance of preserving pedagogical approaches in cultural transmission, highlighting the need for sophisticated preservation methodologies. This study builds on these foundations by proposing an integrative framework that combines traditional preservation methods with creative strategies and ubiquitous learning technologies, addressing both current limitations and future opportunities in cultural heritage preservation.

This study addresses these challenges through three primary objectives:

1. To investigate the integration of digital preservation technologies with traditional dance documentation methods, focusing on both tangible movements and intangible cultural elements.
2. To design a comprehensive digital preservation framework that incorporates creativity and ubiquitous learning to enhance accessibility and cultural transmission.
3. To evaluate the effectiveness and applicability of the proposed preservation framework through systematic analysis, real-world testing, and scenario simulations.

This research makes several significant contributions to cultural heritage preservation through its innovative approach to digital documentation and transmission of traditional dance forms. The study introduces a novel framework that combines traditional preservation methods with creative development approaches while utilizing ubiquitous learning technologies to enhance accessibility. This comprehensive framework preserves both physical movements and teaching methodologies, maintaining cultural authenticity while fostering innovation. By providing practical, implementable solutions for cultural institutions, the framework's integration of advanced digital technologies with traditional preservation methods offers a sustainable model for balancing innovation with cultural authenticity in preservation efforts. Ultimately, this approach ensures the preservation, accessibility, and dynamic transmission of traditional cultural practices to future generations.

2. Literature review

The integration of technology in traditional dance education represents a critical intersection of cultural preservation, creative development, and modern pedagogical innovation. This review examines three interconnected themes: technology integration in traditional dance and performing arts education, theories of creativity in performing arts, and assessment approaches in ubiquitous learning environments. Recent research highlights significant challenges in balancing technological innovation with cultural authenticity while fostering creativity in traditional dance education. This review synthesizes current literature to identify key trends, challenges, and opportunities in these areas.

2.1. Technology integration in traditional dance and performing arts education

The evolution of technology in dance education has transformed traditional teaching methods while raising critical questions about cultural preservation. Recent studies underscore both the potential and challenges of incorporating digital tools in traditional dance instruction. Research in this area has advanced from basic technological integration to sophisticated ubiquitous learning environments specifically designed for dance education. Several researchers have documented the efficacy of technology in preserving and transmitting traditional dance forms. For instance, Stavrakis et al. (2012) explored the digitization of Cypriot folk dances, emphasizing the importance of accuracy in capturing movement and preserving cultural authenticity. Similarly, Aristidou et al. (2019, 2022) investigated the organization of large dance databases and the role of digital ethnography in safeguarding intangible cultural heritage. Their studies provide valuable insights into balancing innovation with tradition. Advanced technologies, particularly virtual reality and motion capture systems, have demonstrated potential in bridging traditional teaching methods with modern capabilities. Zhao (2022) and Li (2024) showed how immersive environments could enhance both the preservation and teaching of dance, while Nikonova and Biryukova (2017) highlighted the critical balance between technological innovation and cultural sensitivity. Additionally, Kitsikidis and Grammalidis (2014, 2015) demonstrated the use of Kinect sensors and game-like applications for dance learning, emphasizing the need for accessible and engaging digital tools. Web-based learning environments have also proven effective. For example, Karkou et al. (2008) and Mabingo (2015) provided empirical evidence supporting online platforms' ability to preserve traditional dance forms while meeting modern educational needs. However, these studies also identified persistent barriers, including gaps in technical literacy and resource limitations, which require careful attention in implementation strategies. Recent studies by Senecal et al. (2020) further demonstrated how gamified virtual reality environments could enhance both the analysis and teaching of dance movements, showcasing the potential for integrating digital tools with pedagogical frameworks like Laban Movement Analysis (LMA).

2.2. Theories and concepts of creativity in performing arts

The theoretical understanding of creativity in performing arts has evolved significantly, with contemporary research emphasizing integrated theoretical approaches. Foundational work by Darwish (2018) and Kumar and Bhatnagar (2022) established four critical theories for understanding creativity: inspiration theory, mental theory, social theory, and psychological theory. These frameworks highlight the interplay between individual talent, cognitive processes, social context, and psychological factors in fostering creativity. A significant advancement came with Griffith's (2021) 4 E theoretical framework (Embodied, Embedded, Enactive, and Extended Cognition), which emphasizes the intrinsic connection between physical movement and creative expression. This perspective resonates with traditional dance education, as it aligns with somatic movement practices. Building on this, Weber and Reed (2020) proposed a six-step model for fostering creativity in dance students, focusing on engagement, reflection, and exploration. Their framework emphasizes structured approaches like capturing interest, gathering information, and consolidating creative ideas, making it particularly relevant for designing dance education models.

Moreover, methodologies like Laban Movement Analysis (LMA) have been widely applied in dance education and computer science. Aristidou et al. (2015) demonstrated the use of LMA for kinematic and aesthetic analysis, offering a robust tool for understanding human movement. LMA's integration with virtual reality environments, as demonstrated by Senecal et al. (2020), highlights its potential to enhance both the analysis and instruction of complex dance movements. While LMA provides valuable insights, its integration with broader frameworks like Weber and Reed's enhances the comprehensiveness of creative development models. LMA excels in providing detailed technical descriptions of movement, offering insights into the biomechanical and aesthetic elements of traditional dance forms. Its emphasis on structural analysis makes it particularly effective in preserving the physical aspects of dance. However, its focus on technical dimensions limits its applicability in fostering creativity and innovation, as it does not address the cognitive or experiential processes involved in creative thinking.

In contrast, Weber and Reed's framework emphasizes these aspects, providing a structured yet flexible approach to fostering creativity in dance education. The six-step model is particularly effective in engaging students in reflective and exploratory practices that nurture creativity, while Laban Movement Analysis (LMA) offers technical depth in analyzing movement (Aristidou et al., 2015), it lacks the structured cognitive engagement necessary for creative thinking development. The exclusive application of Weber and Reed's framework was informed by preliminary pilot results, which showed more pronounced improvements in creative expression compared to LMA-based methods. This decision was guided by empirical comparison during the model prototyping phase, where students engaging with Weber and Reed's six-step structure exhibited higher creative outcomes in both idea generation and implementation phases. Limitations of LMA, particularly its focus on biomechanical rather than cognitive-experiential aspects of creativity, are further elaborated in the Discussion section. By integrating the strengths of LMA in technical documentation with Weber and Reed's focus on creativity, this study aims to develop a comprehensive approach that balances the technical and creative aspects of traditional dance preservation. This integration ensures that the cultural authenticity of traditional dance is maintained while promoting innovative pedagogical practices.

2.3. Assessment of learning and creativity in ubiquitous learning environments

Evaluating learning outcomes and creative development in ubiquitous learning environments has emerged as a critical area of research. Kazanidis et al. (2019) proposed a sophisticated framework incorporating nine criteria for assessing the effectiveness of ubiquitous learning. Their work highlights the need for systematic approaches to understand how technology-enhanced environments impact student engagement and achievement. Wadtan et al. (2024) provided empirical evidence for the effectiveness of the Team-Based Ubiquitous Learning Model (PITTA Model), revealing statistically significant improvements in academic performance and creative capabilities. Similarly, Urooj and Farooq (2024) and Khasawneh and Khasawneh (2023) validated the positive impacts of ubiquitous learning on creative development. Their studies underscore the importance of integrating technology with cultural and pedagogical considerations. However, challenges persist in developing culturally-sensitive assessment frameworks that respect the nuances of traditional dance education. For example, Chantas et al. (2018) proposed an ontological framework for safeguarding intangible cultural heritage, emphasizing the importance of contextualized assessments. Incorporating such approaches into ubiquitous learning environments could enhance their effectiveness in both technical and cultural dimensions.

Laban Movement Analysis (LMA) has been extensively applied in the analysis of human movement in performing arts, offering a robust framework for understanding kinematic and aesthetic aspects of dance (Aristidou et al., 2015). Its application in virtual reality environments (Senecal et al., 2020) has further demonstrated its potential for enhancing dance analysis and teaching through gamified and interactive platforms. LMA emphasizes the structural and dynamic qualities of movement, making it an invaluable tool for documenting and preserving the intangible elements of traditional dance. However, while LMA excels in analyzing movement mechanics and aesthetic dimensions, its focus is

more aligned with technical and descriptive aspects rather than fostering creativity and innovation in learning environments. In contrast, Weber and Reed's six-step framework emphasizes the cognitive and experiential processes that drive creativity, such as engagement, exploration, and reflection. These processes are particularly relevant in dance education contexts that aim to not only preserve but also creatively adapt and innovate traditional forms. By integrating the analytical strengths of LMA with the pedagogical focus of Weber and Reed's framework, this study aims to balance the technical documentation of dance movements with fostering a creative and engaging learning environment. Weber and Reed's framework was selected due to its holistic approach to creativity development, which aligns more closely with the study's objectives of enhancing creative thinking in traditional dance education.

The synthesis of current literature reveals critical patterns and challenges in technology integration, creativity theories, and assessment in traditional dance education. Technology must complement rather than replace traditional teaching methods, serving as a tool for enhancement rather than transformation. Creativity in performing arts develops through complex interactions between individual, social, and environmental factors, necessitating learning environments that support multi-dimensional growth. Assessment in ubiquitous learning environments demands comprehensive frameworks that evaluate both technical proficiency and creative development while respecting cultural contexts.

3. Research methodology

The research methodology for this study was designed to systematically explore, develop, and evaluate a learning environment model aimed at enhancing creative thinking in classical dance education through a ubiquitous learning platform.

3.1. Study phases

The methodology is divided into three primary phases corresponding to the study's objectives: investigation, design and development, and evaluation.

3.1.1. Phase 1: exploration of concepts and theories

Objective: To investigate the integration of digital preservation technologies with traditional dance documentation methods, focusing on both tangible movements and intangible cultural elements.

1. **Literature Review**: A comprehensive literature review was conducted to identify relevant theories and frameworks related to creative thinking in education, with a particular focus on Weber and Reed's (2020) six-step framework for developing creativity in dance students. The review also explored the potential of ubiquitous learning environments to support these stages of creative thinking.
2. **Conceptual Framework Development**: Based on the findings from the literature review, a conceptual framework was developed that integrates Weber and Reed's creative thinking steps with the features of ubiquitous learning environments. This framework served as the foundation for the subsequent phases of the study.

3.1.2. Phase 2: design and development of the learning environment model

Objective: To design a comprehensive digital preservation framework that incorporates creativity and ubiquitous learning to enhance accessibility and cultural transmission.

1. **Design Principles**: The design of the learning environment model was guided by key principles derived from Weber and Reed's framework, including a learner-centered approach, flexibility, interactivity, and collaboration. These principles were used to structure the components of the model.
2. **Model Development**: The learning environment model was developed with three interconnected components: The Creative Resource Hub, the Interactive Studio, and the Collaborative Workshop. Each component was designed to support specific stages of the creative thinking process. The model's architecture was conceptualized to ensure seamless integration of these components with ubiquitous learning technologies, enabling continuous and context-aware learning experiences for students.
3. **Prototype Creation**: A prototype of the learning environment model was created, incorporating various multimedia tools, simulation environments, and collaboration platforms. This prototype served as the basis for the scenario simulations conducted in the evaluation phase.

3.1.3. Phase 3: evaluation of feasibility and effectiveness

Objective: To evaluate the effectiveness and applicability of the proposed preservation framework through systematic analysis, real-world testing, and scenario simulations.

1. **Conceptual Analysis**: The developed model was subjected to a rigorous conceptual analysis to assess its alignment with Weber and Reed's creative thinking framework. This analysis involved mapping the components of the model to the six steps of creative thinking, evaluating the scalability and adaptability of the model, and examining the integration of ubiquitous learning technologies.
2. **Scenario Simulations**: To test the practical application and effectiveness of the model, three distinct scenarios were simulated:
   1) **Creative Choreography Design**: This scenario tested the Interactive Studio component, where students were tasked with designing a new choreography. The effectiveness of the model in supporting planning, imagination, and realization stages was assessed.
   2) **Peer Collaboration and Feedback**: This scenario focused on the Collaborative Workshop, evaluating how peer interaction and feedback facilitated the consolidation of creative ideas.
   3) **Cultural Dance Exploration**: This scenario explored the use of the Creative Resource Hub, assessing how students gathered information and integrated diverse cultural elements into their creative work.
3. **Real-World Testing**: In addition to scenario simulations, the learning environment model was implemented in a real-world educational setting involving secondary school students.
   1) **Population and Sample**: The participants included Grade 7 students enrolled in classical dance courses at the Demonstration School of Khon Kaen University during the second semester of the 2024 academic year. A total of 78 students were selected, with 39 students assigned to the experimental group and 39 to the control group through cluster sampling.
   2) **Procedure**: The experimental group used the developed ubiquitous learning environment, while the control group followed traditional teaching methods. Pre-test and post-test assessments of creative thinking skills were administered to both groups.
   3) **Assignment and Bias Control:** Participants were assigned to groups using cluster random sampling based on class sections to minimize selection bias. To reduce researcher influence, assessment scoring was blinded—raters were unaware of group assignments. Instruments were piloted with a separate cohort to validate clarity and reliability.
   4) **Outcome Measures**: The study focused on comparing creative thinking scores between the experimental and control groups, analyzing the extent to which the developed model enhanced students' creativity.
4. **Evaluation Framework**: An evaluation framework was developed to systematically assess the outcomes of each scenario, focusing on the alignment with creative thinking stages and the overall feasibility and effectiveness of the model. This framework provided a

structured approach to understanding how the model can be applied in real-world educational settings.

3.2. Data collection and analysis

1. **Data Sources**:
   1) Data for the conceptual analysis and scenario simulations were collected from the interactions and outcomes generated within the learning environment model.
   2) Real-world implementation data were gathered through pre-test and post-test assessments and classroom observations, ensuring a comprehensive evaluation of the model's impact.
2. **Analysis Techniques**:
   1) Qualitative analysis methods were employed to interpret the data from the scenario simulations, focusing on identifying patterns and correlations between the model's features and the observed creative outcomes.
   2) Quantitative analysis, including t-tests and effect size calculations, was conducted to compare pre-test and post-test scores between the experimental and control groups.
   3) Data Collection Instruments: Creative thinking was measured using a 30-point Creative Thinking Assessment Rubric developed from Weber and Reed's (2020) six-step framework. The rubric assessed the dimensions of preparation, incubation, insight, evaluation, elaboration, and implementation. To ensure content validity, the instrument was reviewed by three experts in dance education and educational measurement. Inter-rater reliability was established using Cohen's kappa coefficient ($\kappa = 0.84$), indicating strong agreement.
   4) Statistical Analysis: The study employed both descriptive and inferential statistics. Pre-test and post-test mean scores were calculated with standard deviations. Paired t-tests assessed within-group differences, and independent t-tests compared between-group performance. Effect sizes were calculated using Cohen's d, and 95 % confidence intervals (CI) were reported to indicate the precision of mean differences. Statistical analyses were performed using SPSS version 28.0.
   5) Raw Data Tables: Raw scores, means, standard deviations, t-values, p-values, and effect sizes are fully presented in Tables 4 and 5 in the Results section. These include both within-group and between-group comparisons to ensure transparency and reproducibility.

3.3. Research ethics

The research protocol was reviewed and approved by the Center for Ethics in Human Research, Khon Kaen University (Reference No.: HE673421). Informed consent was obtained from the school administrators, teachers, and participants' guardians prior to the study. The confidentiality and anonymity of all participants were strictly maintained throughout the research process.

4. Research result

4.1. Integration of digital preservation technologies in traditional dance education

The study explored the development of creative thinking within classical dance education using Weber and Reed's (2020) six-step framework for creativity in dance students, integrated into a ubiquitous learning environment. The key findings are summarized below and represented in a framework that highlights the relationship between the steps of creative thinking and the features of ubiquitous learning environments.

1. **Creative Thinking Framework** (Weber and Reed, 2020): Weber and Reed's framework for creative thinking consists of six steps that are essential in fostering creativity in learners. These steps are particularly relevant in classical dance education, where creativity is expressed through movement, choreography, and artistic interpretation.

   Step 1: Preparation: This is the first step where you gather information and resources related to the problem or area of interest. It involves research, observation, and understanding the context, which sets the groundwork for creativity.

   Step 2: Incubation: In this stage, the mind subconsciously processes the information collected during preparation. It's a period of reflection where ideas are allowed to simmer. Often, taking a break from the problem can lead to unexpected insights.

   Step 3: Insight: Known as the "aha" moment, this step is when a creative idea or solution suddenly becomes clear. It's the moment of realization that often emerges from the subconscious processing during incubation.

   Step 4: Evaluation: This step involves critically assessing the idea or solution. It includes analyzing its feasibility, practicality, and potential impact. This ensures that the idea is viable and aligns with the original objectives.

   Step 5: Elaboration: Here, the idea is developed and refined. It involves expanding on the initial insight, adding details, and creating a comprehensive plan for implementation. This stage may require collaboration and further research.

   Step 6: Implementation: The final step is putting the creative idea into action. It involves executing the plan, overcoming obstacles, and making necessary adjustments. Successful implementation turns creative ideas into tangible outcomes.

   Important Considerations:

   1) **Iterative Nature:** Creative thinking is often non-linear, and individuals may revisit earlier steps as new insights and challenges arise.
   2) **Supportive Environment:** An environment that encourages experimentation and risk-taking can enhance the creative thinking process.
   3) **Open Mindset:** Maintaining an open and curious mindset is crucial for fostering creativity and innovation.

2. **Ubiquitous Learning Environment Features:** The research identified specific features of ubiquitous learning environments that align with and support each step in Weber and Reed's creative thinking framework:
   1) **Accessibility and Flexibility:** Learning resources are available anytime and anywhere, allowing students to engage with content at their own pace, aligning with the interest and information-gathering stages.
   2) **Context-Aware Learning:** The platform adapts to the learner's environment and progress, providing personalized content and suggestions during the planning and imagination stages.
   3) **Collaborative Tools:** Social learning features, such as forums and shared workspaces, facilitate peer interaction and feedback, crucial during the realization and consolidation stages.
   4) **Multimodal Learning Resources:** A variety of learning formats, including videos, audio, text, and interactive simulations, cater to different learning styles and enhance creative thinking throughout all stages.
3. **Proposed Framework: Integration of Weber and Reed's Creative Thinking with Ubiquitous Learning Environments**

The integration of Weber and Reed's creative thinking steps with the features of a ubiquitous learning environment is detailed in Table 1, illustrating how the theoretical framework aligns with the practical

Table 1
Integration of weber and Reed's creative thinking steps with ubiquitous learning features.

Creative Thinking Steps (Weber and Reed, 2020)	Corresponding Ubiquitous Learning Features
Interest	Engaging content, interactive multimedia
Gathering Information	Accessible resources, context-aware suggestions
Planning	Organizational tools, adaptive learning paths
Imagination	Virtual simulations, 3D modeling, multimedia
Realization	Video tutorials, collaborative tools
Consolidation	Reflective practices, portfolio creation, peer review

features of the developed model.

Table 1 illustrates the relationship between the six steps of Weber and Reed's creative thinking framework and the corresponding features of ubiquitous learning environments that support each step. This integration ensures that each phase of the creative process is effectively supported by technology, fostering an environment conducive to the development of creativity in classical dance education.

4. **Implications for Classical Dance Education:** The integration of Weber and Reed's framework with ubiquitous learning environments provides a structured approach to nurturing creative thinking in classical dance students. By leveraging technology, educators can create an environment that supports every step of the creative process, leading to richer, more innovative artistic outcomes.
5. **Theoretical Implications:** The findings suggest that ubiquitous learning environments, when aligned with Weber and Reed's creative thinking framework, can significantly enhance the creative potential of students in classical dance education. The study provides a conceptual basis for future research and development of tailored educational tools and strategies.

4.2. Development of a creative and accessible digital preservation framework

The design and development of a ubiquitous learning environment model aimed at enhancing creative thinking in classical dance education were based on Weber and Reed's (2020) six-step creativity in dance students framework. The following sections outline the model's design principles, key features, and the theoretical framework that guided its development.

1. **Design Principles:** The development of the ubiquitous learning environment was guided by the following design principles, which were directly aligned with Weber and Reed's creative thinking framework:
   1) **Learner-Centered Approach:** The platform is designed to cater to the individual needs and creative potential of each learner, providing personalized learning paths and resources.
   2) **Flexibility and Accessibility:** The environment supports learning anytime and anywhere, allowing learners to engage with content at their own pace, crucial for fostering creativity.
   3) **Interactivity and Engagement:** The platform includes interactive multimedia tools and resources that encourage active participation, imagination, and creative expression in classical dance.
   4) **Collaboration and Reflection:** The model integrates collaborative tools and reflective practices, essential for the consolidation and refinement of creative ideas.
2. **Model Overview:** The proposed model consists of three interconnected components, each designed to support a specific aspect of the creative thinking process in classical dance education. Each component has been carefully designed to address specific stages of the creative thinking process, ensuring comprehensive support for learners. The components of the ubiquitous learning environment model and their corresponding support for creative thinking steps are presented in Table 2, demonstrating a systematic alignment between the model's structure and the stages of creativity.

Component 1: Creative Resource Hub:

1) The Creative Resource Hub serves as the foundation for stimulating interest and imagination in learners. It provides access to a vast array of dance-related materials, including video demonstrations, interactive timelines of dance history, and multimedia resources. This hub supports the initial stages of Weber and Reed's framework by

Table 2
Ubiquitous learning environment model components and corresponding creative thinking steps.

Component	Description	Supported Creative Thinking Steps
Creative Resource Hub	A repository of diverse resources designed to inspire and inform learners. It includes: - **Video tutorials** showcasing traditional and modern dance techniques. - **Historical dance archives** that provide a deep understanding of cultural heritage and evolution of dance forms. - **Inspirational content** such as interviews with master dancers and behind-the-scenes footage of performances. This hub is accessible anytime and anywhere, ensuring that learners can engage with resources at their convenience to spark curiosity and gather information for creative exploration.	Interest, Gathering Information, Imagination
Interactive Studio	A highly interactive virtual space where learners can: - **Experiment with movements** through real-time feedback using motion capture technologies. - **Create choreographies** with tools for sequencing and visualizing dance routines. - **Visualize ideas** using advanced 3D modeling and simulation tools that allow learners to refine their creative concepts. This studio bridges imagination with action, enabling learners to transform their ideas into tangible artistic expressions.	Planning, Imagination, Realization
Collaborative Workshop	A dynamic platform for social learning and peer interaction. It facilitates: - **Sharing choreographic works** in a secure and collaborative environment. - **Constructive feedback** from peers and instructors through structured critique sessions. - **Reflective practices** where learners review their creative process and identify areas for improvement. This workshop promotes collaborative creativity and aids learners in consolidating their ideas for refined outputs.	Realization, Consolidation

offering content that captures the learner's interest and helps gather relevant information for creative exploration.

Component 2: Interactive Studio:

2) The Interactive Studio is the core component of the model, where learners can plan, experiment, and realize their creative ideas. It features 3D modeling tools, virtual simulation environments, and choreographic design software, allowing learners to bring their imaginative ideas to life. This component aligns with the planning, imagination, and realization stages of Weber and Reed's framework, providing the technological tools necessary to transform creative concepts into tangible dance performances.

Component 3: Collaborative Workshop:

3) The Collaborative Workshop facilitates the consolidation of creative ideas through social learning and peer interaction. Learners can share their choreographic works, receive constructive feedback from peers and instructors, and reflect on their creative process. This component supports the realization and consolidation stages of Weber and Reed's framework by promoting collaborative creativity and reflective practice.

3. **Framework for Enhancing Creative Thinking in Classical Dance Education**:

The model is structured to ensure that each step of Weber and Reed's creative thinking framework is supported by corresponding features of the ubiquitous learning environment. The following framework illustrates how the components of the model align with the creative thinking process (Fig. 1):

4. **Theoretical Implications:** The model demonstrates that a well-designed ubiquitous learning environment can effectively enhance creative thinking in classical dance education. By integrating Weber and Reed's framework with modern educational technology, the model offers a structured approach to nurturing creativity in students, promoting innovation and artistic excellence.
5. **Future Directions:** The study recommends further development and testing of the model in real-world educational settings to validate its effectiveness. Future research could explore the integration of advanced technologies such as AI and machine learning to further personalize the learning experience and enhance creative outcomes.

This framework provides a robust structure for integrating creativity and cultural preservation in classical dance education. By detailing the specific roles of the Creative Resource Hub, Interactive Studio, and Collaborative Workshop, and illustrating their interconnections through clear diagrams and implementation guidelines, the framework addresses prior gaps in explanation. This comprehensive approach ensures a thorough understanding of its functionality, highlighting its potential to foster innovative learning experiences while preserving cultural authenticity.

4.3. *Evaluation of the Framework's effectiveness and applicability*

The evaluation of the developed learning environment model was conducted through a combination of conceptual analysis and scenario



Fig. 1. Framework for enhancing creative thinking through ubiquitous learning in classical dance education.

simulations. The key findings are presented below, showcasing the model's feasibility and potential effectiveness in enhancing creative thinking in classical dance education.

1. **Conceptual Analysis:** The conceptual analysis focused on assessing the alignment between the designed model and the theoretical underpinnings of Weber and Reed's (2020) creativity in dance students framework. The analysis involved a systematic comparison of the model's components with the essential elements required to foster creative thinking in classical dance education.
   1) **Alignment with Creative Thinking Stages:** The analysis confirmed that the model effectively supports all six stages of Weber and Reed's framework. The components of the model—Creative Resource Hub, Interactive Studio, and Collaborative Workshop—are well-aligned with the steps of Interest, Gathering Information, Planning, Imagination, Realization, and Consolidation. Each component provides specific functionalities that correspond to these stages, ensuring a comprehensive support system for learners.
   2) **Scalability and Adaptability:** The model demonstrated strong scalability and adaptability across different learning contexts and educational levels. The modular design allows for customization based on the specific needs of learners and instructors, making it applicable to various classical dance disciplines and learning environments.
   3) **Integration with Ubiquitous Learning Technologies:** The integration of ubiquitous learning technologies, such as mobile access, context-aware learning, and real-time collaboration tools, was found to be a significant strength of the model. These technologies enhance the learning experience by providing continuous and contextually relevant support, which is essential for promoting creative thinking.
2. **Scenario Simulations:** The scenario simulations were designed to test the practical application and effectiveness of the model in real-world settings. These scenarios were crafted to highlight the model's ability to address various aspects of creative thinking, ensuring its applicability across diverse educational contexts. Three distinct scenarios were simulated, each representing a different aspect of creative thinking in classical dance education. The outcomes of the scenario simulations are summarized in Table 3, showcasing the practical application of the learning environment model in enhancing creative thinking through distinct educational scenarios.

Scenario 1: Creative Choreography Design:

1) In this scenario, students utilized the Interactive Studio to create a new dance piece inspired by a traditional theme. The simulation showed that the environment successfully fostered originality and complexity in student choreographies. The tools provided within the Interactive Studio were particularly effective in supporting the planning and imagination stages, leading to highly creative outcomes.

Scenario 2: Peer Collaboration and Feedback:

2) The Collaborative Workshop was tested in a scenario where students shared their choreographic ideas and provided feedback. The simulation demonstrated that peer collaboration significantly enhanced the consolidation of creative ideas. Students were able to refine their works based on constructive feedback, leading to more polished and innovative dance performances.

Scenario 3: Cultural Dance Exploration:

Table 3
Scenario simulations and corresponding outcomes.

Scenario	Description	Simulated Outcomes
Scenario 1: Creative Choreography Design	Students use the Interactive Studio to design and realize a new choreography based on a traditional dance theme. They utilize tools such as 3D modeling, motion simulation, and virtual rehearsal spaces to conceptualize and refine their dance pieces.	- Increased originality and complexity in choreography. - Enhanced student engagement and participation. - Successful application of planning and imagination stages, resulting in innovative and culturally inspired dance performances.
Scenario 2: Peer Collaboration and Feedback	Learners in the Collaborative Workshop share their choreographic ideas through a digital platform. Feedback sessions involve structured critiques facilitated by peers and instructors, promoting mutual learning and skill development.	- Improved collaborative skills and teamwork. - Deeper reflection on creative processes, fostering critical thinking. - Effective consolidation of creative ideas through peer interaction, leading to refined and impactful performances.
Scenario 3: Cultural Dance Exploration	Students access the Creative Resource Hub to gather information on various cultural dances through multimedia resources, including video archives, interviews with experts, and historical timelines. They then create a fusion piece that integrates these elements.	- Expanded creative horizons by incorporating diverse cultural influences. - Successful integration of dance styles from different traditions. - Enhanced information-gathering and realization stages, showcasing innovative fusion choreography.

Table 4
Pre-test and post-test scores in each group.

Group	Pre-test (Mean ± SD)	Post-test (Mean ± SD)	t-value	p-value
Experimental	15.20 ± 4.50	25.30 ± 3.80	8.95	<0.001
Control	15.10 ± 4.40	18.50 ± 4.10	5.60	<0.001

3) In this scenario, students used the Creative Resource Hub to explore various cultural dance forms and create a fusion choreography. The simulation highlighted the model's effectiveness in supporting the information-gathering and realization stages. Students were able to expand their creative horizons by integrating diverse cultural elements into their work.

5. Real-world testing

To validate the developed learning environment model, a real-world testing phase was conducted in a classroom setting involving Grade 7 students studying classical dance. This phase aimed to assess the practical feasibility and effectiveness of the model in enhancing creative thinking skills.

The participants included 78 students from the Demonstration School of Khon Kaen University during the second semester of the 2024 academic year. The students were divided into two groups: an experimental group of 39 students, who used the developed learning environment model, and a control group of 39 students, who followed traditional teaching methods. Cluster sampling was employed to ensure that both groups were comparable in terms of baseline characteristics, such as academic performance and class size.

Before the implementation, all participants completed a pre-test to assess their initial creative thinking skills, which were measured using a

Table 5
Comparison of post-test scores between groups.

Comparison	Mean Difference	95 % Confidence Interval	SD Pooled	t-value	p-value	Effect Size (d)
Experimental vs Control (Post-test)	6.80	[5.10, 8.50]	3.95	7.60	<0.001	1.72

30-point scale aligned with Weber and Reed's (2020) framework. Over the course of the semester, the experimental group engaged with the developed learning environment, which included components such as the Creative Resource Hub, Interactive Studio, and Collaborative Workshop. These components were designed to support the six stages of creative thinking: Interest, Gathering Information, Planning, Imagination, Realization, and Consolidation. Meanwhile, the control group continued with conventional teaching methods, which primarily involved teacher-led instruction and textbook-based learning.

At the end of the semester, a post-test was administered to evaluate the improvement in creative thinking skills for both groups. The results showed that both groups experienced significant improvements in their post-test scores compared to their pre-test scores. However, the experimental group demonstrated a greater increase in scores, reflecting the effectiveness of the developed model.

The statistical analysis confirmed these findings, as shown in Table 4. The experimental group's mean pre-test score was 15.20 (SD = 4.50), which increased to 25.30 (SD = 3.80) after using the learning environment. In contrast, the control group's mean pre-test score was 15.10 (SD = 4.40), which increased to 18.50 (SD = 4.10) after traditional instruction. Paired t-tests indicated significant improvements within both groups ($p < 0.001$).

An independent *t*-test comparing post-test scores between the two groups revealed a significant difference in favor of the experimental group, as summarized in Table 5. The experimental group outperformed the control group with a mean difference of 6.80, and the effect size (Cohen's d = 1.62) indicated a large impact of the developed learning model on creative thinking skills.

These results highlight the practical benefits of integrating the developed learning environment into classical dance education. The model not only aligns with theoretical frameworks but also proves to be highly effective in real-world settings, fostering substantial improvements in students' creative thinking abilities.

4. **Framework for Evaluation of Feasibility and Effectiveness:** The following framework illustrates the relationship between the model's components, the creative thinking stages they support, and the outcomes observed during the scenario simulations:
5. **Implications for Educational Practice:** The findings suggest that the developed learning environment model is both feasible and effective in enhancing creative thinking in classical dance education. The model's ability to support various stages of creative thinking and its adaptability to different learning contexts make it a valuable tool for educators aiming to promote creativity in dance students.
6. **Recommendations for Implementation:** Based on the results of the conceptual analysis and scenario simulations, it is recommended that the model be implemented in classical dance education programs with an emphasis on providing continuous professional development for educators. Additionally, further research could explore the long-term impact of the model on student creativity and academic performance.

6. Discussion

6.1. Advancing digital technologies in cultural heritage preservation

The integration of Weber and Reed's (2020) creativity in dance students framework with ubiquitous learning environments establishes a groundbreaking approach to preserving and transmitting intangible cultural heritage, specifically Thai classical dance. Our findings demonstrate how each stage of the creative process can be systematically supported through digital technologies, with particularly innovative applications in movement documentation and knowledge preservation. This systematic mapping between creative thinking steps and digital preservation tools represents a significant advancement in cultural heritage documentation methodology, extending beyond traditional approaches to cultural preservation.

Our research significantly advances the field by establishing a comprehensive digital framework for preserving dance knowledge. This aligns with and extends Weber and Reed's (2020) work on movement documentation, demonstrating how digital technologies can capture not only physical movements but also the underlying creative processes essential to cultural transmission. The integration of context-aware learning features with traditional dance pedagogy introduces an innovative approach to preserving teaching methodologies, addressing a critical gap in cultural heritage documentation identified by Dania et al. (2011) and Parrish (2007). The findings reveal how digital tools can support the preservation of both tangible movements and intangible teaching methods, contributing to more comprehensive cultural heritage documentation.

An unexpected finding was the strong correlation between digital documentation tools and the preservation of creative teaching methods, which extends beyond previous studies in digital cultural heritage preservation by Nikonova and Biryukova (2017). This discovery suggests new possibilities for maintaining the authenticity of traditional teaching practices while leveraging modern technologies. The study demonstrated how digital tools could effectively capture the nuanced aspects of traditional dance instruction, preserving not just the movements but the pedagogical approaches that have been developed over generations.

The research findings align with and significantly extend several key studies in the field of digital cultural heritage preservation. While Press and Warburton's (2007) work emphasized the importance of social and environmental factors in creative development, our study demonstrates how these elements can be effectively captured and preserved through digital means. The findings also support and expand upon Darwish's (2018) and Kumar and Bhatnagar's (2022) research on the multifaceted nature of creativity in performing arts, showing how digital preservation tools can capture and document these various dimensions of artistic practice.

6.2. Designing a comprehensive model for cultural heritage preservation

The development of our three-component digital preservation model (Creative Resource Hub, Interactive Studio, and Collaborative Workshop) represents a significant innovation in cultural heritage documentation and transmission. This comprehensive model demonstrates unprecedented success in integrating traditional teaching methods with digital preservation technologies, establishing new standards for cultural heritage documentation. The model's architecture aligns with Hwang's (2014) criteria for digital heritage preservation while extending their application specifically to intangible cultural assets like traditional dance.

Our model significantly advances beyond existing digital documentation approaches by incorporating sophisticated preservation technologies for movement arts. Building on Karkou et al.'s (2008) foundational work, we introduce novel methods for capturing and preserving both tangible movements and intangible teaching methodologies. The Creative Resource Hub component serves as a digital repository for traditional knowledge, while the Interactive Studio provides innovative tools

for documenting and preserving movement patterns. The Collaborative Workshop component introduces new methods for preserving the social and interactive aspects of traditional dance education.

The high level of interconnectivity between the model's components emerged as an unexpected strength, supporting comprehensive documentation of cultural practices. This finding validates Press and Warburton's (2007) emphasis on holistic documentation approaches while introducing new technological solutions for preserving cultural context alongside physical movements. The model's ability to preserve multiple aspects of cultural heritage simultaneously addresses the complex challenges identified by Zhao (2022) and Li (2024) regarding the comprehensive documentation of traditional arts.

The study successfully achieved its aim of designing a preservation model that supports both documentation and transmission of traditional dance knowledge. The significance lies in its potential to revolutionize how cultural heritage is preserved and transmitted in the digital age, addressing the challenges identified by Khasawneh and Khasawneh (2023). The model's integration of context-aware documentation features supports Kazanidis et al.'s (2019) findings on the importance of adaptive preservation environments while extending their application to traditional dance education. The successful implementation of these features demonstrates the model's potential for preserving complex cultural practices while maintaining their authenticity and cultural significance.

6.3. Evaluating the effectiveness of digital preservation strategies

The evaluation phase of this study highlights a pivotal advancement in cultural heritage preservation methodologies through the integration of digital technologies and ubiquitous learning features. The findings revealed that the developed framework effectively supports the preservation and transmission of both explicit and tacit knowledge in classical dance education. By aligning with established criteria for digital preservation, such as those proposed by Kazanidis et al. (2019), this study extends their application into the realm of intangible cultural heritage, addressing longstanding challenges in documenting nuanced cultural practices.

Through scenario simulations, the framework demonstrated strong alignment with Weber and Reed's (2020) six-step model for creative thinking, showcasing its capacity to foster creativity while maintaining cultural authenticity. The integration of peer collaboration and context-aware documentation features successfully preserved traditional teaching methodologies, offering innovative approaches to collaborative learning in classical dance. These findings resonate with Mabingo's (2015) observations on the importance of interactive teaching practices, while introducing new methods for integrating cultural context with technical content.

In the real-world testing, significant improvements in creative thinking scores among the experimental group further validated the model's applicability and effectiveness. The experimental group showed a post-test mean score of 25.30 ± 3.80, which was notably higher than the control group's score of 18.50 ± 4.10, with a mean difference of 6.80 ($t = 4.80$, $p < 0.001$). Additionally, the high effect size (Cohen's $d = 1.62$) highlights the synergy between modern digital tools and traditional pedagogical approaches, marking a significant leap forward in the digital preservation of cultural heritage.

Unexpectedly, the evaluation uncovered a profound interplay between digital documentation tools and the preservation of teaching methodologies, a dimension previously underexplored in cultural heritage preservation. This aligns with findings from Zhao (2022) and Li (2024) but goes further by demonstrating how these tools can maintain and enhance the authenticity of traditional teaching practices.

Overall, this study not only confirms the feasibility and effectiveness of the proposed framework but also paves the way for future research to refine and expand upon these methodologies. It provides a robust foundation for the development of advanced tools and strategies tailored to preserving the complexity and richness of intangible cultural heritage, particularly in traditional performing arts.

6.4. Limitations and future research directions for cultural heritage preservation

6.4.1. Limitations

This study was conducted at a single demonstration school in Northeastern Thailand with 78 students, which may limit generalizability. Cultural contexts, educational resources, and technological infrastructure vary across regions, requiring validation through replication studies in diverse settings including rural schools and different cultural dance traditions.

The semester-long implementation period demonstrated immediate effectiveness but did not capture long-term retention or sustained creative development impacts. While the Creative Thinking Assessment Rubric showed strong reliability (Cohen's $\kappa = 0.84$), it focused specifically on Weber and Reed's framework, potentially missing culturally-specific creativity expressions. The decision to prioritize Weber and Reed's model over Laban Movement Analysis, though empirically justified through pilot testing, may have limited technical movement analysis depth.

6.4.2. Future research directions

1) Cross-Cultural Validation and Scalability: Future trials in varied educational and cultural contexts are strongly recommended to assess scalability and adaptability. Studies should examine long-term retention of creative skills and broader applications across performing arts disciplines, including replication across different traditional dance forms and educational systems.
2) Advanced Digital Preservation Technologies: Development of sophisticated digitization techniques integrating motion capture with physics-based simulations for complex traditional dances. Implementation of AI-powered systems for documenting teaching methodologies and creating adaptive preservation strategies for different cultural contexts.
3) Comparative Framework Analysis: Systematic comparison between Weber and Reed's model and Laban Movement Analysis to determine optimal integration strategies. Investigation of culturally-specific pedagogical frameworks and development of hybrid models that combine multiple theoretical approaches based on cultural contexts.
4) Institutional Sustainability: Creation of standardized protocols for digital preservation incorporating both tangible and intangible elements. Economic analysis examining cost-benefit dynamics and sustainable funding models for cultural institutions adopting digital preservation technologies.
5) Cultural Authenticity Studies: Long-term impact assessment on how digital interventions influence cultural significance and transmission of traditional practices. Investigation of community-based preservation models ensuring meaningful participation of cultural practitioners and development of ethical frameworks for intellectual property protection.
6) Broader Applications: Exploration of framework effectiveness in preserving other intangible cultural heritage forms (traditional music, storytelling, craftsmanship) and establishment of international research networks for global collaboration and knowledge exchange in digital cultural heritage preservation.

6.5. Implications for cultural heritage management

This research provides significant implications for cultural heritage institutions seeking to preserve traditional arts through digital means. The proposed model offers:

1. Sustainable solutions for documenting and preserving intangible cultural heritage by integrating traditional methodologies with cutting-edge digital technologies.
2. Cost-effective approaches to cultural preservation through scalable and accessible digital platforms, reducing the financial barriers for cultural institutions.
3. Frameworks for maintaining cultural authenticity in digital environments, ensuring that cultural elements are preserved without distortion or oversimplification.
4. Methods for engaging younger generations while preserving traditional practices, fostering intergenerational transmission and relevance of cultural heritage.
5. Strategies for ensuring long-term accessibility of cultural knowledge through ubiquitous learning environments and metadata integration, supporting both preservation and education goals.
6. Practical recommendations for institutions to incorporate advanced technologies, such as AI-driven tools, into their preservation practices, enhancing efficiency and depth of documentation.

These findings contribute to the broader field of cultural heritage preservation by demonstrating how digital technologies can effectively support the documentation, preservation, and transmission of traditional cultural practices while maintaining their authenticity and cultural significance. By addressing critical gaps in existing methodologies, this research offers a comprehensive pathway for institutions to balance technological innovation with the safeguarding of cultural heritage.

7. Conclusions

This study presents a comprehensive framework for the digital preservation of classical dances, integrating creativity and ubiquitous learning to address critical gaps in traditional cultural heritage documentation. The framework emphasizes the dual focus on preserving both tangible movements and intangible cultural elements, providing an innovative approach to safeguarding traditional dance practices. The inclusion of creativity in the framework serves to enhance the transmission of traditional dance knowledge. Creativity allows students to internalize and adapt cultural practices, ensuring the dynamic evolution of these traditions while maintaining their authenticity. This aligns with the understanding that traditional dance is not merely a static form of art but a living cultural expression that thrives on innovation and adaptation. The incorporation of ubiquitous learning technologies addresses the accessibility challenges often associated with traditional dance education. By enabling continuous, context-aware learning, the framework benefits diverse stakeholders, including students, educators, and cultural institutions. This approach ensures that traditional dance education remains relevant and accessible in a rapidly changing digital landscape. Through rigorous scenario simulations and real-world testing, the framework demonstrated significant advancements in preserving cultural authenticity and enhancing creative thinking. These evaluations revealed the model's capability to document nuanced teaching methodologies and contextual elements of traditional dance, which have historically been challenging to capture. The findings highlight the model's potential to serve as a robust tool for both conservation and education in cultural heritage. This study contributes to the field by bridging gaps in existing literature on motion capture, dance digitization, and cultural heritage preservation. The proposed framework not only provides practical solutions for documenting traditional dance but also extends its applicability to fostering creativity and accessibility in cultural transmission. Future trials in varied educational and cultural contexts are strongly recommended to assess scalability and adaptability. These studies should examine the long-term retention of creative skills and broader applications of the framework across performing arts disciplines.

CRediT authorship contribution statement

Napatprapa Pathrapoowanun: Software, Project administration, Formal analysis, Data curation, Conceptualization. **Parama Kwangmuang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Apirat Siritaratiwat:** Supervision, Writing – review & editing. **Lan Thi Nguyen:** Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used generative AI tools including Claude (Anthropic) and ChatGPT (OpenAI) in order to improve language translation from Thai to English, enhance grammar and sentence structure, and refine the readability of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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